

Centrifugal Gas Compressor - Maintenance Manual

(Excerpt)

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Section 4: Routine Maintenance

This manual excerpt covers Type CGC-3 centrifugal gas compressors deployed across PETRONAS Upstream platforms and Gas & New Energy LNG complexes.

Routine maintenance comprises three cadences: daily walkdown, monthly preventive maintenance, and annual major inspection.

4.1 Daily Walkdown

Operations staff shall perform a daily walkdown of each compressor unit. The walkdown checklist includes: visual inspection of seals for leaks, vibration spot-check at bearings, lubrication oil level check, and verification that suction and discharge pressures are within normal operating envelope.

Findings shall be recorded in the Site Operations Log and abnormal conditions raised as a maintenance work order in the CMMS.

4.2 Monthly Preventive Maintenance

Monthly preventive maintenance covers: replacement of suction filter elements, sampling of lubrication oil for laboratory analysis, inspection of inlet guide vane actuators, and recalibration of vibration probes.

Estimated downtime: 4 hours. The work shall be scheduled in the Plant Maintenance Plan to coincide with low demand periods.

4.3 Annual Major Inspection

The annual major inspection requires the unit to be isolated and locked out following the site Permit-to-Work procedure. Activities include: rotor borescopy, replacement of dry-gas seals, full lubrication system flush, and re-alignment of the driver-driven train.

Estimated downtime: 5 days. The inspection shall be performed by a qualified rotating equipment specialist with Type CGC-3 endorsement.

Section 5: Failure Modes

Surge: Recognised by rapid pressure oscillation and audible bark. Operators shall reduce load and open the recycle valve to recover. Sustained surge may damage the rotor and shall be reported within 1 hour.

High Vibration: A vibration trip is configured at 7.1 mm/s peak. On trip, the unit shall not be restarted until the root cause is identified.

Seal Gas Leak: Indicated by rising secondary seal vent flow. The unit shall be shut down within 4 hours and the seal cartridge replaced.

Section 6: Reporting

All maintenance events shall be logged against the `asset_id` in the Group Asset Register and surfaced to the `operations_daily` table via the `maintenance_event_count` column at the silver layer.

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